The Heat Price of Big Data

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Paper: The Heat Price of Big Data: Leveraging Machine Learning to Analyze the Urban Heat Effects of Data Center Alley in Washington D.C.'s Hinterlands

Motivation

- Urban Heat Island effect is becoming more intense every year
  - Land Surface Temperature is a key indicator (although this is not the same as UHI)
- Emergence of the “Data Center Alley” (Ashburn, VA)
- Environmental Safety vs. Technological Progress



amazon



Google



Microsoft

Background Information

- Various approaches to quantify UHI
  - Most studies use LST instead of air temperature because of the amount of data available.
- Surface Layer Heat Index corresponds to the LST component of UHI
  - Canopy Layer Heat Index (near surface air temperature) corresponds to the air temperature, which is experienced by humans. This is not considered as previously stated.
- Many features of Urban environments correlate with LST
  - Some ways of determining the features include street typology, LULC, etc.
- UHI effects have notable fluctuations depending on seasonality (Acosta et. al, 2024).

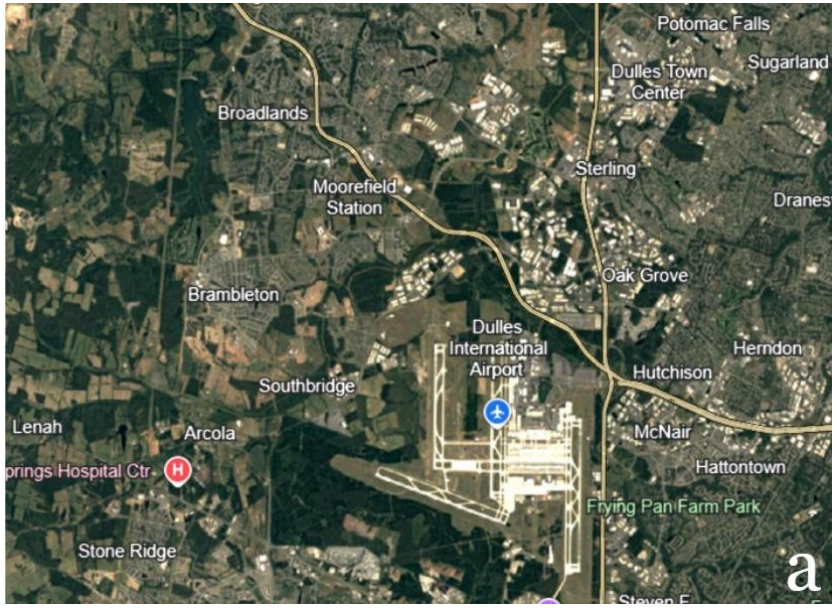
Study Area

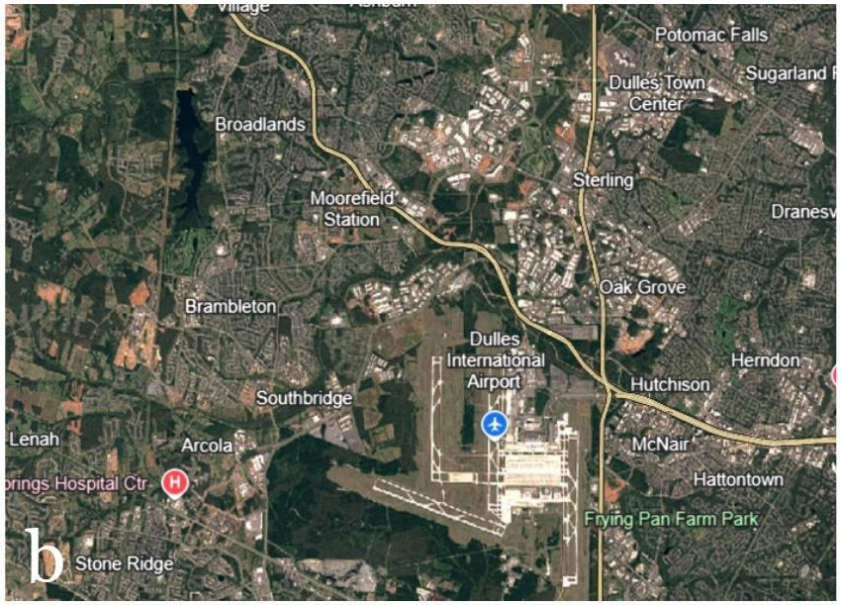
The study area is Ashburn, a satellite city to Washington DC.

Recently, it has been named the “Data Center Alley” as a reference to the Silicon Valley.

70% of the world’s internet traffic goes through Ashburn. (Datacenters.com, 2019)

The top image is from 2010; the bottom image is from 2020.

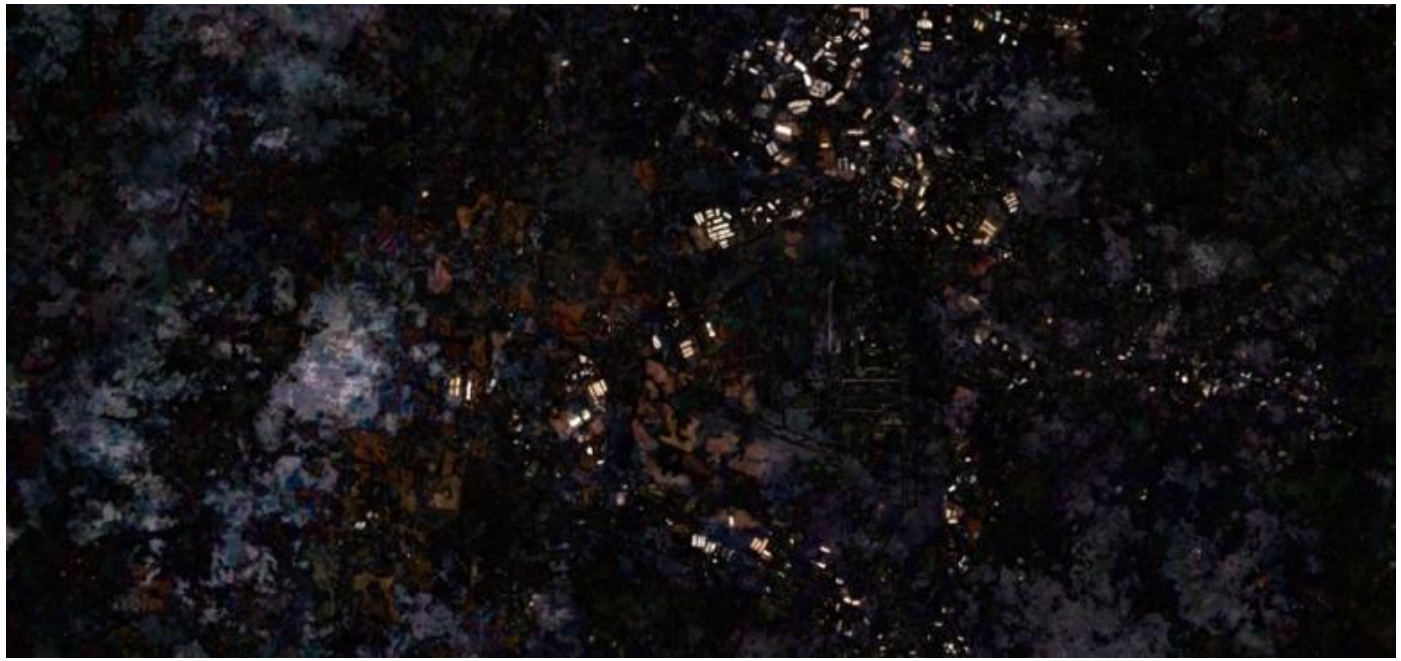




Recent Urbanization

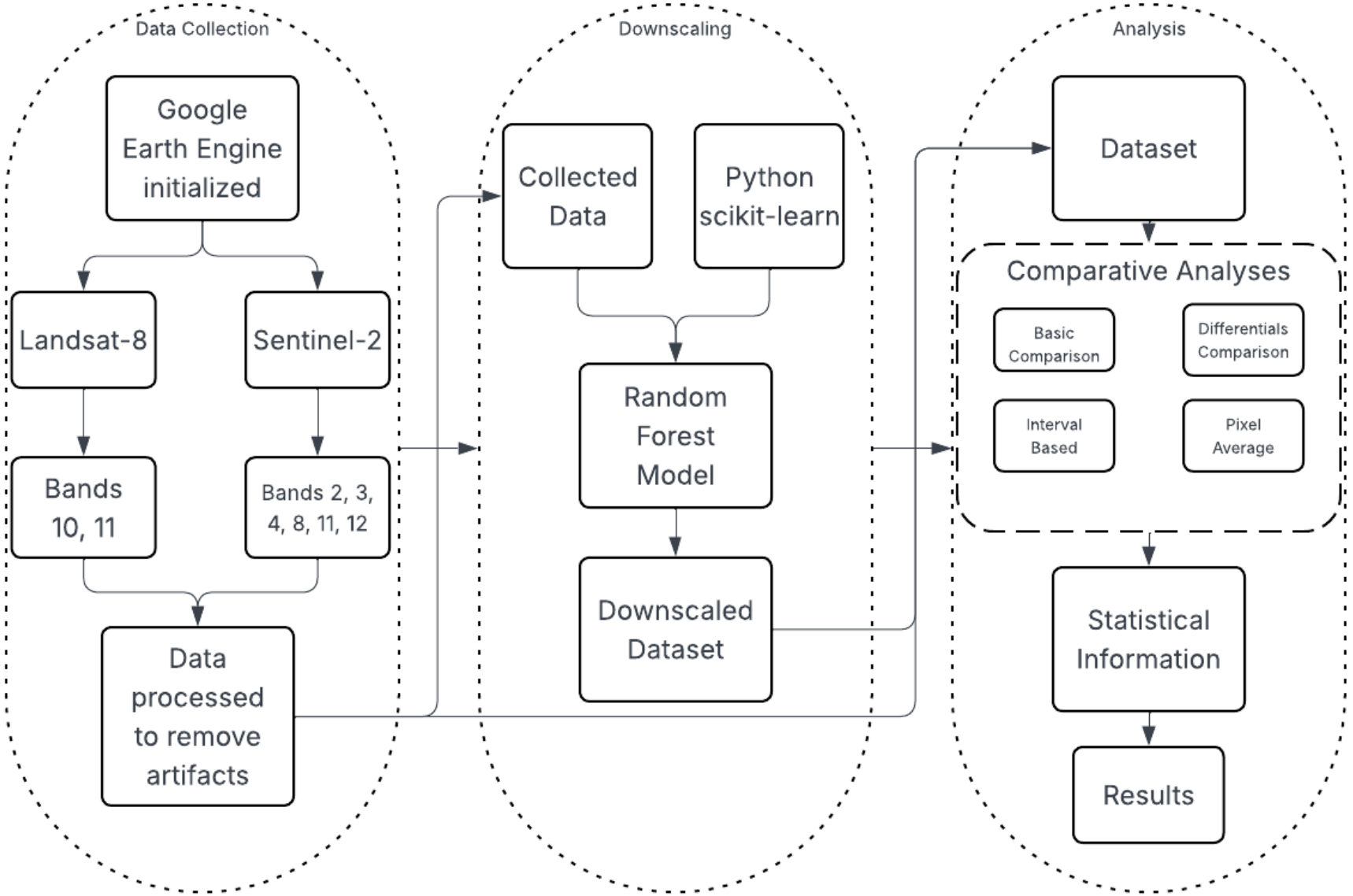
In recent years, Ashburn has urbanized far more.

The white rectangles seen are all new buildings built in the last decade. These include data centers.



Has the rise of the Data Center Alley itself led to a significant increase in land surface temperature?

Methodology Overview

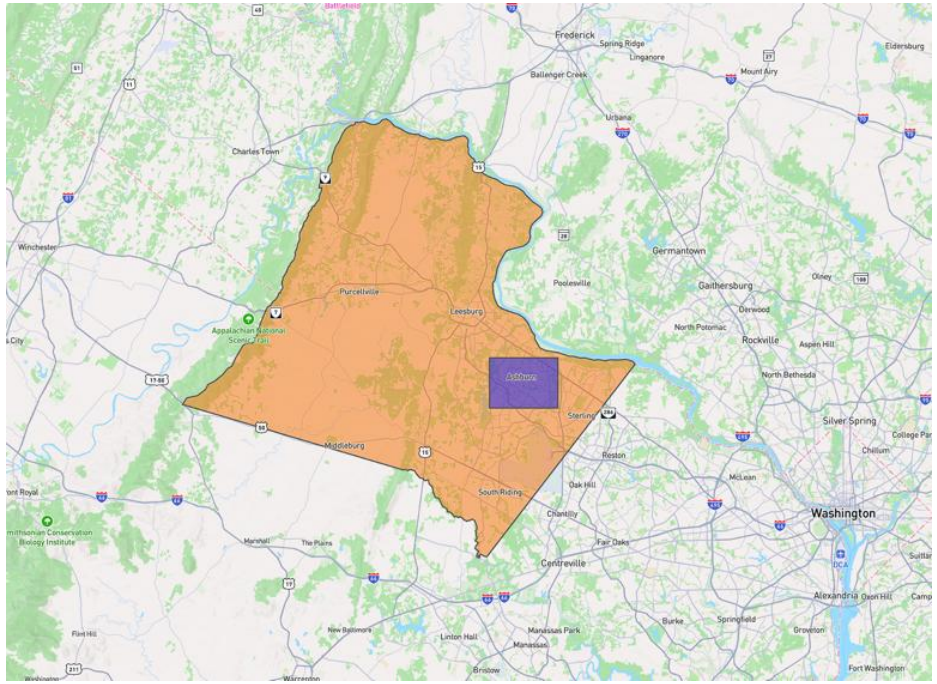


Data Collection

- Google Earth Engine (EE) was utilized via a Python script
- LST data collected from Landsat-8's Thermal Bands (Bands 10 and 11)
- Data required for downscaling is taken from Sentinel-2 (specifically, Band 2, Band 3, Band 4, Band 8, Band 11, and Band 12)
- Data is clipped for the region of interest and processed to account for artifacts (i.e., clouds)

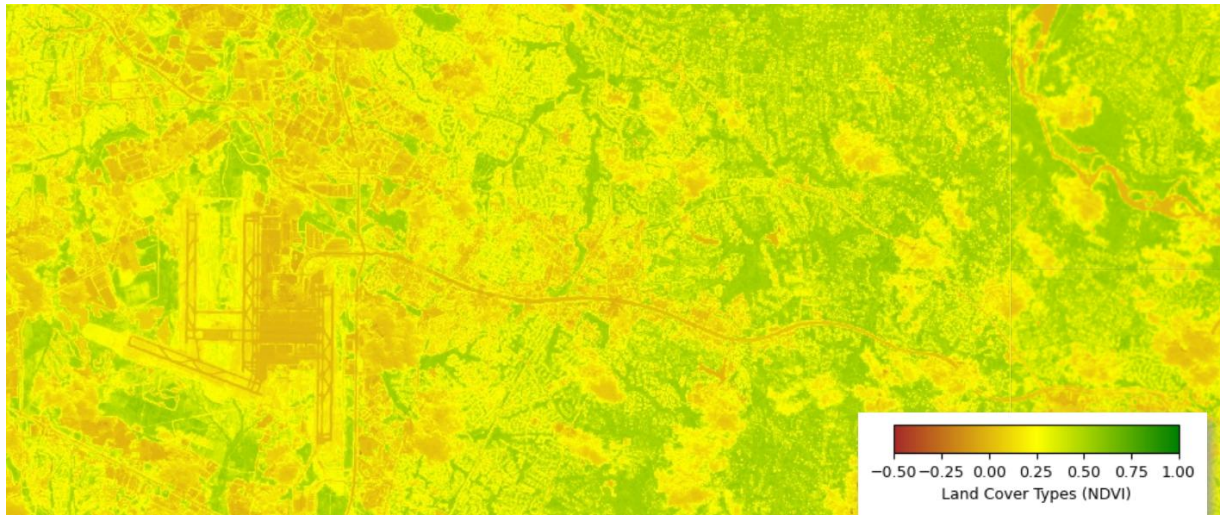
Specific LST Selection

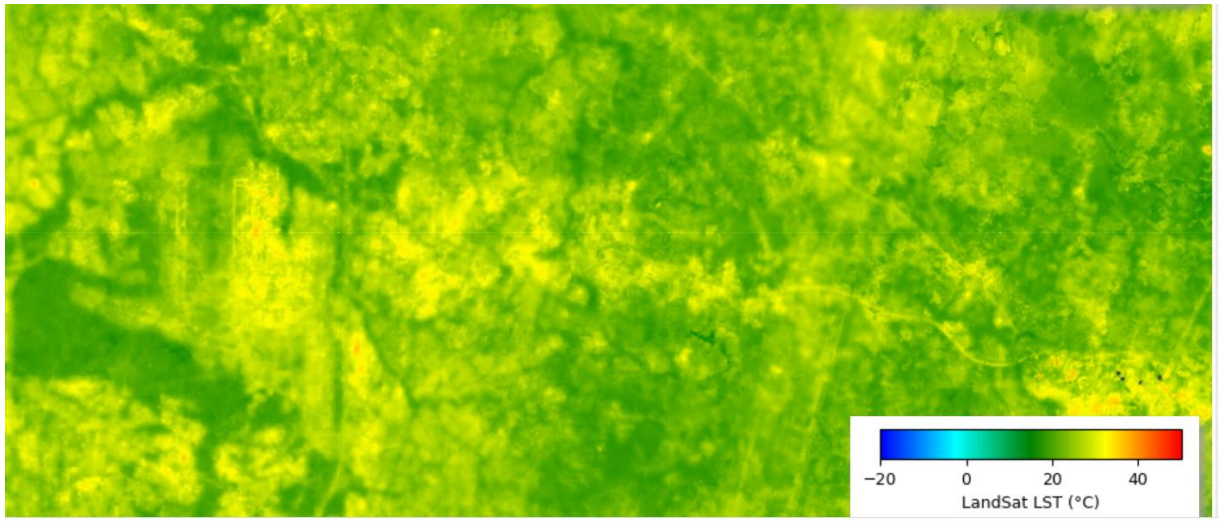
- Dates in late August were selected
- Dates were vetted to have similar weather conditions
- Several different regions of interest were considered



Downscaling Motivation

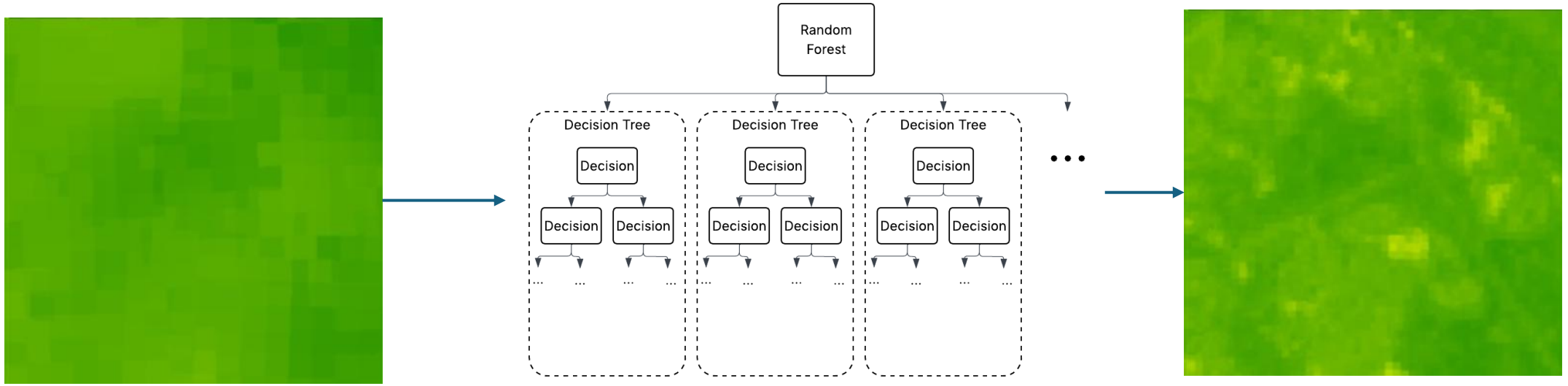
- High correlation between Normalized Difference Vegetation Index (NDVI), Normalized Difference Water Index (NDWI), Normalized Difference Building Index (NDBI), and LST data. (Aboutalebi, 2024)
- NDVI, NDWI, and NDBI (10 m) often come in better resolution than LST data (100 m).





Downscaling Process

- Random forest regressor model was used to downscale LST data using NDVI, NDWI, and NDBI.
- The model comes from the Python “scikit-learn” library
- Tests ran with both 100 and 200 trees for downscaling.

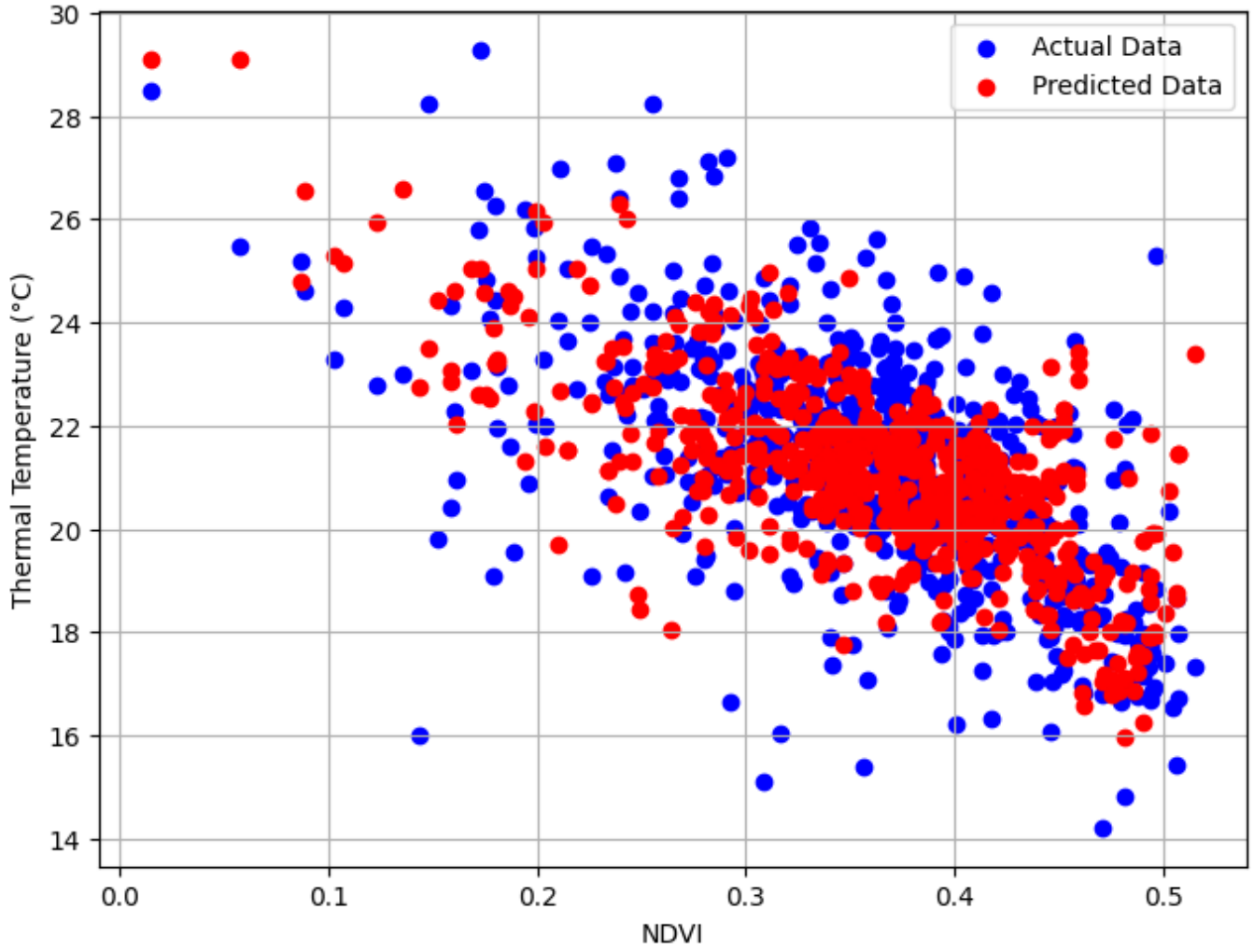


Accuracy of Downscaling

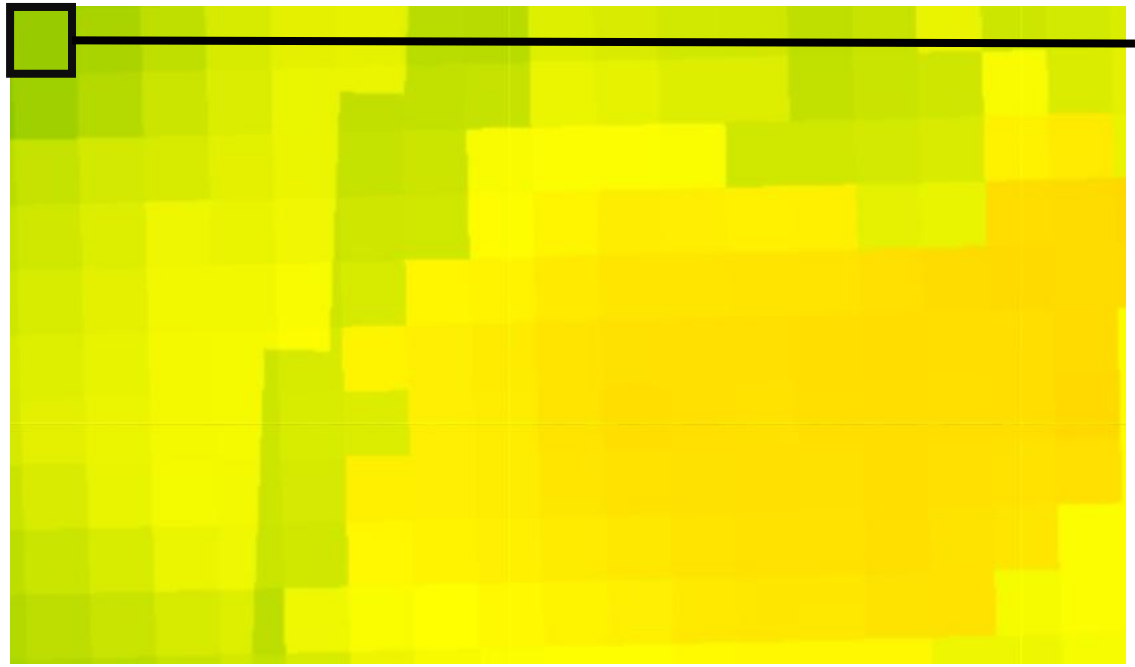
Downscaling predictions shown.

Actual data is usually more varied with more “extreme points”.

For this reason, both downscaled and raw LST data were analyzed to capture the full picture.



Mathematical Background



$$T_{\text{Date}}(x, y) = \text{Temperature on Date at x, y}$$ [1]

Basic Comparative Analysis $\{T_{Date}(x, y) \mid (x, y) \in \text{ROI}\}$ [2]

Differential Analysis $\{\Delta_{Date}[T](x, y) \mid (x, y) \in \text{ROI}\}$ [2, 3]

Point-Based Analysis $\{\frac{\sum_{d \in \text{DateInterval}} T_d(x, y)}{|\text{DateInterval}|} \mid (x, y) \in \text{ROI}\}$ [2, 4]

Interval Analysis $\{\frac{\sum_{(x,y) \in \text{ROI}} T_d(x, y)}{|\text{ROI}|} \mid d \in \text{DateInterval}\}$ [2, 4]

Let $\bar{T}(t) = \frac{\sum_{(x,y) \in \text{ROI}} T_t(x, y)}{|\text{ROI}|}$. The study needs to find the behavior of $\frac{d\bar{T}}{dt}$. This is one collapse of the 4-dimensional function *T*, and it serves as a good indicator as computing the exact behavior is not useful due to data variability. These approaches are data heavy, hence the downscaling.

[1]Temperature is converted from original data (scaled Kelvin) into Celsius
[2]These denote *multi-sets*.
[3]This is with respect to *Date*, and step size *h* = 1 year.
[4]This could also be replaced with a median function.

Mathematical Approaches

Analysis Approach	Derivative Estimate Approach
$\{T_{Date}(x,y) \mid (x,y) \in \mathrm{ROI}\}$	Gives exact values of $T_{date}(x,y)$. Can be seen as an (informal!) approach using $\Delta_{Date,h}[T](x,y)$ for large step sizes h. Taking $\lim_{h\to 0} \frac{\Delta_{Date,h}[T](x,y)}{h}$ may yield $\frac{\partial T}{\partial t}$ where t = Date.
$\{\Delta_{Date}[T](x,y) \mid (x,y) \in \mathrm{ROI}\}$	Directly uses $\Delta_{Date,h}[T](x,y)$ with h = 1 year (or perhaps some other time interval). May also be seen as a second derivative approach.
$\{\frac{\sum_{d\in \mathrm{DateInterval}} T_d(x,y)}{\lvert\mathrm{DateInterval}\rvert} \mid (x,y) \in \mathrm{ROI}\}$	Treats an interval as a single date by taking the mean/median. Uses approach 1 again.
$\{\frac{\sum_{(x,y)\in \mathrm{ROI}} T_d(x,y)}{\lvert\mathrm{ROI}\rvert} \mid d \in \mathrm{DateInterval}\}$	Tries to compute $\bar{T}$. One may estimate the finite difference from this.

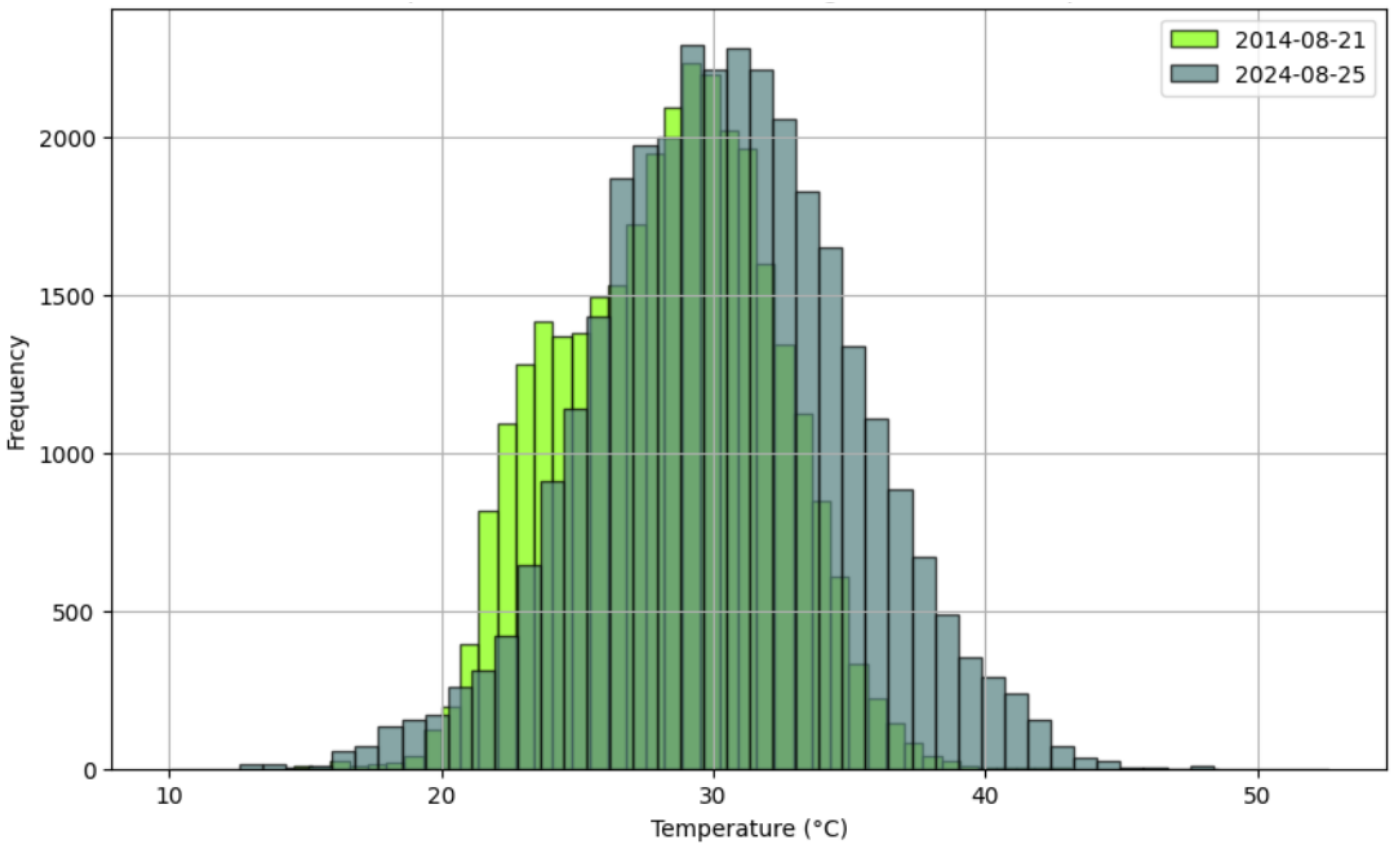
$$\frac{d\bar{T}}{dt} \approx \frac{\partial T}{\partial t}^{*}$$

*It should be noted that this is somewhat an abuse of notation.

Central LST Analysis

The figure demonstrates one of the many results of this study. It compares raw LST data between two different days from 2014 and 2024.

Independent sample t-test suggests that the increase is significant (with $p < 0.01$).



$$\{T_{Date}(x, y) \mid (x, y) \in \text{ROI}\}$$

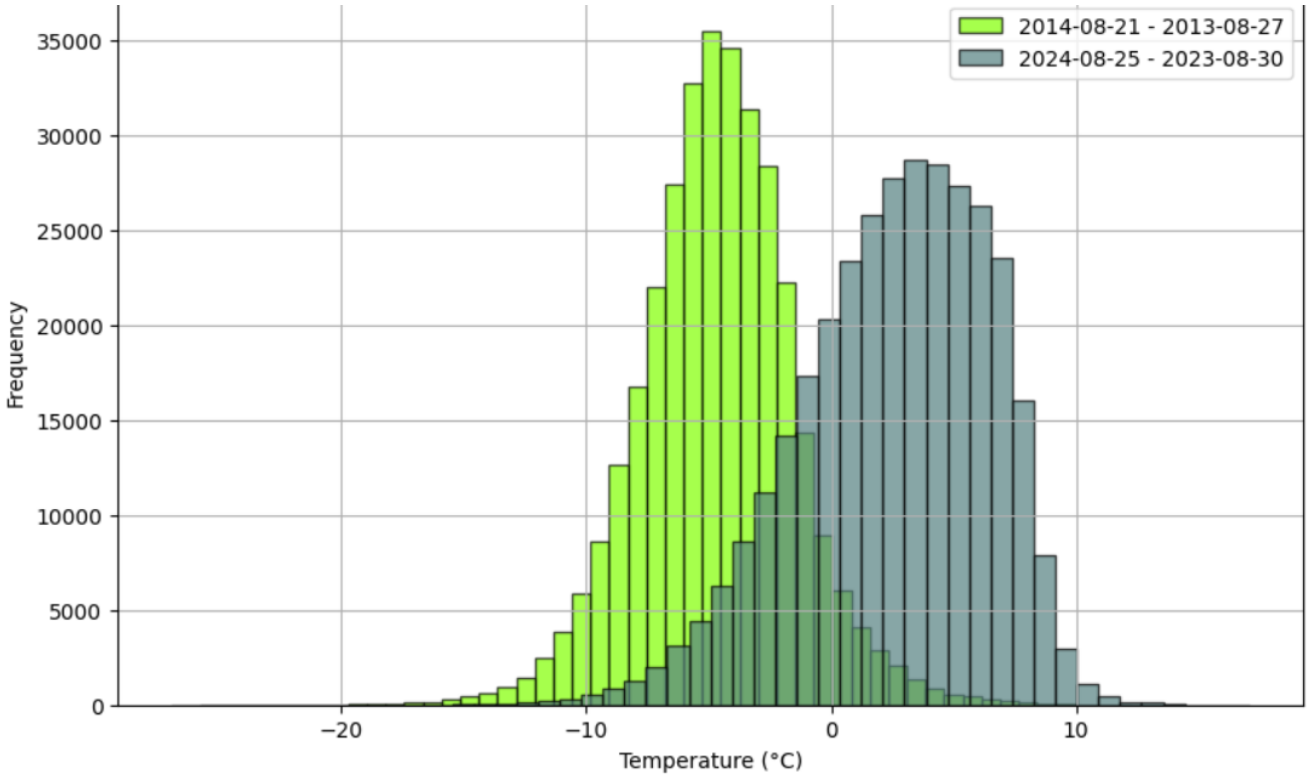
Other Analysis

Many other analyses were performed. This one demonstrates the growth (or difference) in temperature from 2013-2014 and 2023-2024 (on similar days).

Other analyses not shown in this presentation include considering long time intervals using medians or averages. Additionally, these tests ran on different regions of interests and time intervals (as well as downscaled data).

All these tests support that LST has drastically increased in areas with heavy urbanization.

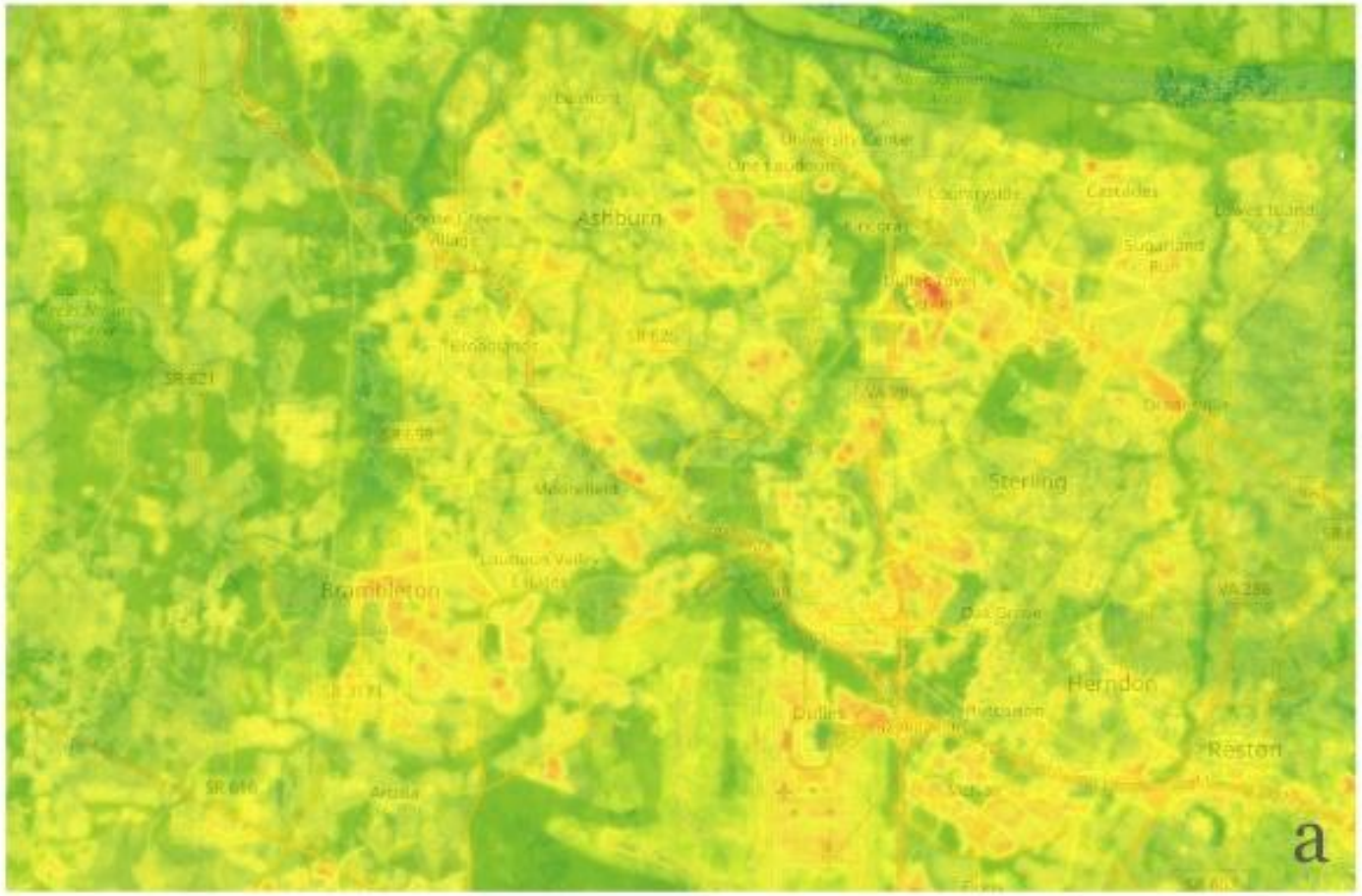
Code is online. (Zhang, 2025)*



$$\{\Delta_{Date}[T](x,y) \mid (x,y) \in \mathrm{ROI}\}$$

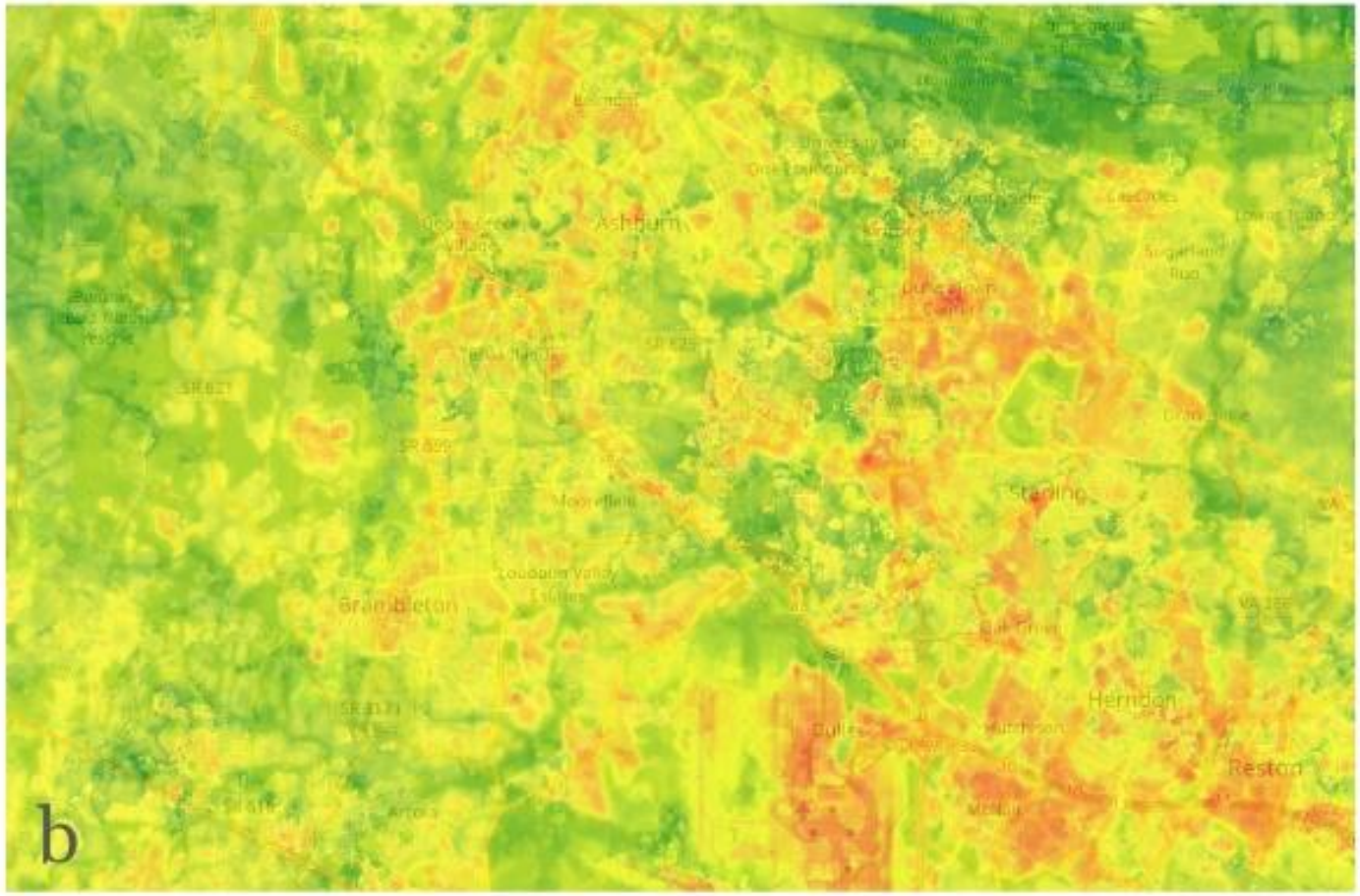
*Available at: https://doi.org/10.7910/DVN/IPPGC

LST Map of Ashburn (2014)



(This is the same data from central LST analysis but visualized on a map)

LST Map of Ashburn (2024)



Implications

- LST of Ashburn has heavily increased recently, especially in areas with many data centers
- Heavy expansion of urban heat islands
- Other features of an urban area also play an effect
  - Examples: commuters to D.C., more houses and shopping centers being built, Dulles Airport, etc.

Future Work

- As time continues, more data will be available for study
  - Crucial for running more advanced statistical tests or getting more consistent results
- Better ML models can fix some of the issues of the random forest predictions
  - These models may also identify causes or correlations that have been previously unnoticed
- Application of this study onto CLHI or air temperature.
- Investigations into the solutions for this problem

Responsible Urban Development

- Industries should be more careful and aware of the environmental stress put on the Ashburn region
- Implement more environmentally friendly strategies to mitigate effects (e.g., green architecture)
- Irresponsible handling will not just affect local areas, but global environmental health

Thank you!

Questions